

ASSIST DEVICES FOR MOBILITY

UNIT-5

Cranes Crutches, and Walkers

Cranes are assistive walking devices that provide limited support to patients who have minor balance or strength impairments. They function by increasing the base of support and redistributing weight from the lower limbs.

Types of Canes

- **Standard (single-point) cane:** Provides the least support; used for mild balance issues.
- **Offset cane:** Designed to center the user's weight over the shaft, suitable for those with hand weakness.
- **Quad cane:** Has a four-legged base that offers more stability, commonly used by patients recovering from stroke.

Indications

- Early-stage osteoarthritis
- Minor hemiplegia
- Post-orthopedic surgery recovery

Crutches Crutches allow for partial or full unloading of one or both lower limbs and are used for more significant impairments than canes.

Types of Crutches

- **Axillary crutches:** Commonly used, fit under the armpit. Suitable for temporary use.
- **Forearm (Lofstrand) crutches:** Cuff around the forearm allows greater mobility and less strain under arms. Suitable for long-term use.

Gait Patterns with Crutches

- Two-point gait
- Three-point gait
- Four-point gait
- Swing-to and swing-through gaits

Walkers provide the most stability among walking aids. They are used by individuals with poor balance, weakness, or coordination.

Types of Walkers

- **Standard walker:** Requires lifting to move; suitable for frail individuals.
- **Two-wheeled walker:** Front wheels allow forward rolling.
- **Rollator:** Four-wheeled walker with brakes and a seat.

Applications

- Elderly with generalized weakness
- Severe arthritis
- Post-operative rehabilitation

Wheeled Mobility

- Wheeled mobility includes assistive devices that enable individuals with mobility impairments to move around efficiently, both indoors and outdoors. These devices are vital for independence, rehabilitation, and quality of life improvement. They can be broadly classified into manual and powered systems.

1. Manual Wheelchairs

Manual wheelchairs are propelled by the user or a caregiver. They are most commonly used due to their lightweight, ease of transportation, and low maintenance.

Types

- **Standard Wheelchair:** Heavier and basic; suitable for short-term use.
- **Lightweight/Ultra-lightweight Wheelchair:** Easier propulsion; used for long-term users.
- **Sports Wheelchair:** Designed for activities like basketball or tennis.

Design Components

- **Push rims, rear wheels, casters, brakes, folding frame, seat and back upholstery.**

2. Powered Wheelchairs

Powered wheelchairs use electric motors and batteries, allowing users to move independently without physical effort.

Control Interfaces

- Joystick
- Sip-and-puff (for high-level SCI patients)
- Head-array and chin controls

Drive Configurations

- **Rear-Wheel Drive:** Stable at high speed, good outdoors.
- **Mid-Wheel Drive:** Tight turning radius, ideal for indoor use.
- **Front-Wheel Drive:** Maneuvers well over uneven terrain.

Seating, Positioning, and Pressure Management

Seating Goals:

- Prevent pressure ulcers and skin breakdown by distributing pressure evenly.
- Maintain neutral spinal alignment to support posture and reduce musculoskeletal stress.
- Enhance comfort, function, and endurance during prolonged sitting, especially for individuals with limited mobility.

Importance of Proper Seating and Positioning:

- Poor seating can lead to deformities, respiratory issues, and digestive problems.
- Proper posture promotes optimal biomechanical alignment, aids in respiration, and improves upper limb function for mobility and activities of daily living.

Cushion Types and Their Characteristics

- **Foam Cushions:**
 - Lightweight and inexpensive.
 - Suitable for users with low to moderate risk of pressure sores.
 - Compress over time and may lose effectiveness.
- **Gel Cushions:**
 - Provide excellent pressure distribution.
 - Can be combined with foam bases for better stability.
 - Require regular maintenance to prevent gel migration.
- **Air Cushions:**
 - Use air-filled cells (e.g., ROHO cushions) to adjust pressure.
 - Ideal for high-risk individuals with limited sensation.
 - Require frequent monitoring for proper inflation.
- **Hybrid Cushions:**
 - Combine foam, gel, and/or air technologies for customized support.

Positioning Aids and Their Role

- **Lateral Supports:** Maintain trunk alignment and prevent leaning.
- **Headrests:** Support the cervical spine, especially in high-level SCI.
- **Pelvic Belts:** Secure pelvis in a neutral position, reducing posterior pelvic tilt.
- **Medial Knee Thigh Supports and Footrests:** Prevent leg adduction and support proper lower limb alignment.

Pressure Mapping Technology:

- Uses sensor mats to measure pressure distribution at the seating interface.
- Visual heat maps identify areas of high pressure that may require adjustment.

Pressure Management Techniques:

- Regular repositioning (e.g., every 2 hours for immobile patients).
- Tilt-in-space and recline features in wheelchairs to offload pressure.
- Incorporating functional reach and push-up maneuvers for pressure relief.

Device Prescription and Clinical Decision-Making

Steps:

1. Patient Assessment
2. Functional Needs Evaluation
3. Environmental Context Review
4. Device Trials
5. Prescription and Follow-up

Remote setting for wheel chair

Smart Features:

- GPS navigation
- Fall detection
- IoT pressure monitoring

Tele-rehabilitation:

- Online therapy sessions
- Remote caregiver control

Adaptation for Sports

- Adaptation for sports involves modifying mobility assistive devices—especially wheelchairs—to suit the unique physical and functional demands of sports participation among individuals with disabilities. These adaptations enhance **performance, safety, comfort, and accessibility**, enabling inclusive sports participation.

Wheelchair Sports

Wheelchair sports include:

- **Wheelchair Basketball** – Involves quick turns, passes, and propulsion, requiring high agility.
- **Wheelchair Tennis** – Players use specially designed chairs with enhanced maneuverability and larger front wheels.
- **Wheelchair Racing** – Involves aerodynamic chairs with hand rims optimized for propulsion speed.
- **Rugby, Fencing, and Table Tennis** – All involve highly customized chairs for sport-specific movement patterns.

Design Modifications for Sports Wheelchairs

- **Cambered Wheels:** Angled wheels provide greater lateral stability, reduce tipping, and improve turning speed.
- **Rigid Frame:** Eliminates moving parts for higher speed and responsiveness.
- **Anti-Tip Bars:** Prevent backward falls during aggressive maneuvers.
- **Lightweight Materials:** Use of aluminum, titanium, or carbon fiber to reduce mass and increase speed.
- **Adjustable Backrest and Seat:** Enhances ergonomic support depending on sport demands.

Benefits of Sports Adaptation in Assistive Devices

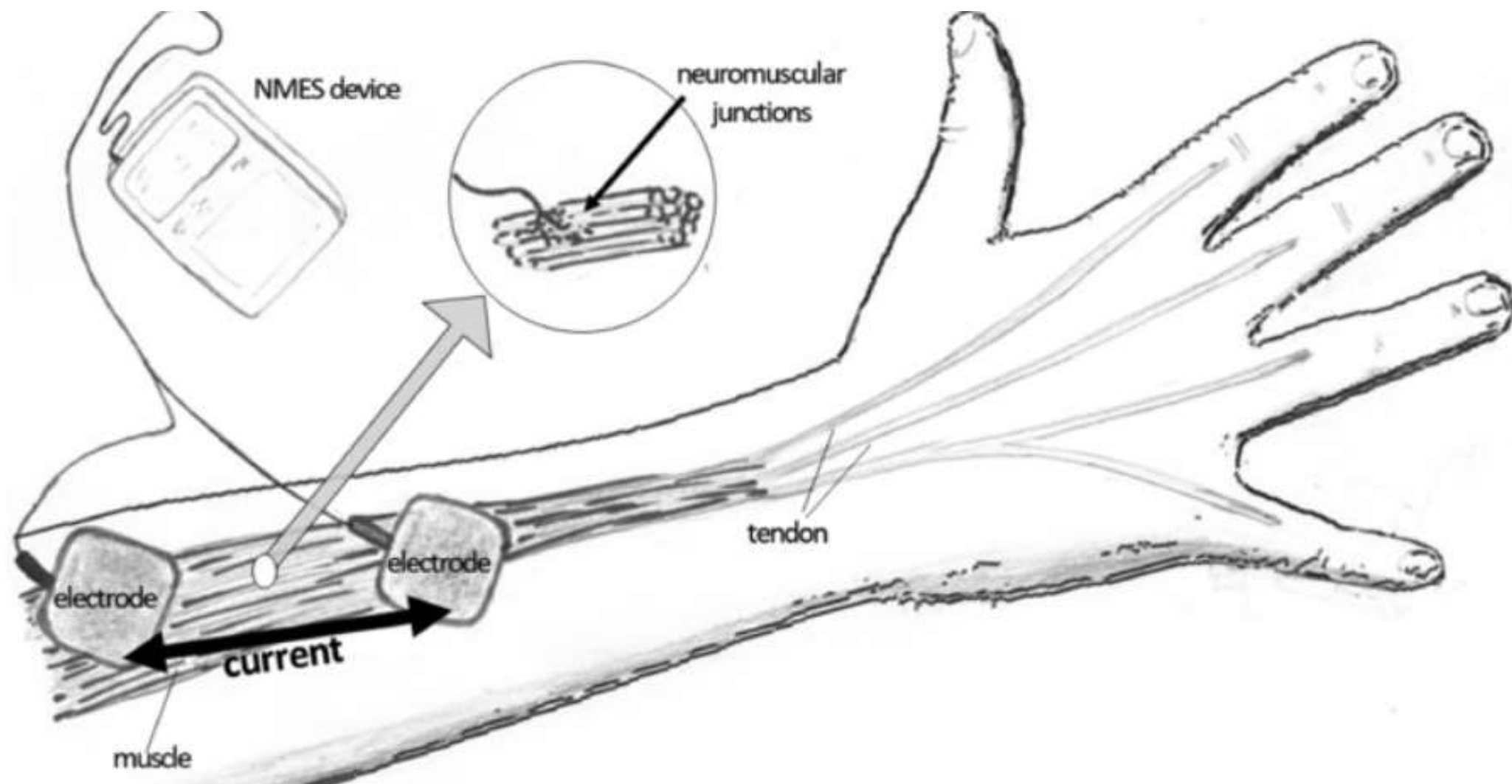
- Promotes **physical fitness** and **mental health**.
- Enables **competitive participation** and **social inclusion**.
- Encourages **innovation** in assistive device design.

Neuromuscular Electrical Stimulation (NMES)

- **Neuromuscular Electrical Stimulation (NMES)** is a therapeutic technique that uses electrical impulses to evoke muscle contractions by stimulating motor nerves. It is primarily used to improve muscle strength, enhance motor control, and aid mobility in individuals with neuromuscular impairments.

Principle

- NMES works by applying surface electrodes over targeted muscle groups. These electrodes deliver low-frequency electrical pulses, mimicking the natural action potential from the central nervous system, thus causing muscle contraction.



Applications in Mobility Assistance

1. Muscle Re-education:

Helps patients regain voluntary control over paralyzed or weakened muscles (e.g., post-stroke or spinal cord injury).

2. Prevention of Muscle Atrophy:

Regular stimulation maintains muscle mass in immobilized patients.

3. Enhancement of Gait:

Used in foot drop correction and walking assistance via Functional Electrical Stimulation (FES), a subset of NMES.

4. Improving Blood Circulation:

Muscle contractions promote venous return, reducing the risk of deep vein thrombosis (DVT).

5. Pain Reduction and Spasticity Management:

Provides therapeutic benefits by relaxing spasms and reducing discomfort.

Common Devices:

- **FES Systems** – Often used in gait training.
- **Portable NMES Units** – Used in clinical or home settings for rehabilitation.

Advantages

- Non-invasive
- Improves motor function
- Customizable for individual needs

Restoration of Motor Function

NMES facilitates the **re-education of muscles** by artificially inducing contractions in paralyzed or weakened muscles. This helps patients relearn muscle control after conditions such as:

- Stroke
- Spinal cord injury
- Traumatic brain injury
- It promotes **neuroplasticity**, enabling the brain and spinal cord to reorganize and form new neural connections.

Gait Improvement and Walking Assistance

NMES, particularly in the form of **Functional Electrical Stimulation (FES)**, enhances gait in individuals with **foot drop** or partial paralysis. By stimulating the **peroneal nerve** during walking, NMES improves:

- Foot clearance
- Balance and coordination
- Walking speed and endurance
- This contributes significantly to **independent ambulation**.

Muscle Strengthening and Atrophy Prevention

In immobilized patients (e.g., post-surgery or injury), NMES:

- Preserves muscle mass
- Prevents disuse atrophy
- Improves muscle tone and endurance
- These outcomes are crucial for maintaining physical functionality during recovery.

Spasticity and Pain Management

NMES is effective in **reducing spasticity** by inducing reciprocal inhibition in spastic muscles and promoting relaxation. Additionally, it:

- Alleviates chronic pain
- Improves joint mobility and posture

Sports Rehabilitation and Functional Training

In athletes or active individuals recovering from injury, NMES:

- Aids in early rehabilitation
- Shortens recovery time
- Reinforces strength and neuromuscular control

Wheeled mobility solution for a participant at the Paralympics

- Wheeled mobility solutions—such as sports wheelchairs—play a transformative role in the performance, independence, and quality of life of Paralympic athletes. These devices are **not just assistive aids but high-performance equipment** tailored to optimize competitive ability in various sporting disciplines. The implications can be examined from the following perspectives:

1. Enhanced Athletic Performance

Custom-built sports wheelchairs are:

- **Lightweight and aerodynamic**, improving speed and maneuverability.
- **Sport-specific** (e.g., basketball, tennis, racing), offering improved biomechanics for precise control.
- Designed for **maximum power transfer**, enabling rapid acceleration and turns.

2. Individualized Fit and Functionality

Wheelchairs used by Paralympic athletes are **custom-engineered** based on:

- Body size, shape, and posture
- Specific impairments and strengths
- Sport-specific rules and requirements

Technical and Biomechanical Advantages

Wheeled mobility solutions offer **biomechanical optimization**, including:

- Optimal seating posture for **efficient propulsion**
- Suspension systems to reduce **impact stress**
- Enhanced **braking and turning systems** for safety

Economic and Technological Considerations

However, there are some challenges:

- High-performance wheelchairs are **costly**, and access may be limited for athletes from low-resource settings.
- Continuous **technological upgrades** are needed to remain competitive, necessitating sponsorship or institutional support.

Exoskeletal Assist Device for Mobility

- An **exoskeletal assist device**, or **exoskeleton**, is a wearable, external mechanical structure designed to support or enhance a person's physical mobility. These devices are typically powered (using electric motors or hydraulics) and work in tandem with the user's movements to assist walking or standing, especially for individuals with neuromuscular impairments or spinal cord injuries.



Types of Exoskeletons

1. Passive Exoskeletons

1. Do not use motors or power sources.
2. Use springs or dampers to redistribute loads and assist movement.
3. Common in industrial settings to reduce strain.

2. Active (Powered) Exoskeletons

1. Use electric motors, batteries, or hydraulic actuators.
2. Offer powered assistance for walking, standing, or lifting.
3. Used in medical rehabilitation and personal mobility.

Key Components

- **Frame:** Lightweight material (aluminum, carbon fiber) to support the body.
- **Motors/Actuators:** Provide joint movement (e.g., hips, knees, ankles).
- **Sensors:** Detect user intention and body position (e.g., gyroscopes, accelerometers).
- **Control System:** Uses algorithms to synchronize movement with the user's efforts.
- **Power Supply:** Rechargeable batteries provide energy to the motors.
- **User Interface:** May include joystick, wrist remote, or voice command for control.

Applications in Mobility

1.Rehabilitation

1. Used in **physiotherapy** for stroke or spinal cord injury patients.
2. Facilitates **relearning of gait patterns** by enabling repetitive, supported walking.
3. Provides **real-time feedback** for clinicians and patients.

2.Personal Assistance

1. Helps **paraplegic individuals** perform activities like standing and walking.
2. Enhances **independence and quality of life** at home or in the community.

3.Military and Industrial Use (non-clinical)

1. Reduces fatigue and injury risk during prolonged physical activity.
2. Assists in carrying heavy loads or maintaining posture.

Advantages

- Improves **gait and posture** in individuals with mobility impairments.
- Reduces **muscle atrophy and bone loss** associated with immobility.
- Enhances **cardiovascular health** through regular supported movement.
- Promotes **neuroplasticity** during rehabilitation.

Exoskeletal assist devices are a breakthrough in biomedical engineering, blending robotics and human movement science to **restore mobility, independence, and dignity**.